



Learning a Language Through Reading: A Meta-analysis of Studies on the Effects of Extensive Reading on Second and Foreign Language Learning

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Abstract

Research on extensive reading (ER) has quite consistently highlighted its impact on second/foreign language learning. However, it remains unclear what the exact components of ER are that make it profitable. In practice, there appears to be variation in the application of key elements of ER, and programs vary in the kinds of additions they make to these elements. The aim of this meta-analysis was to assess the impact of ER on second/foreign language learning and to examine which key elements and additions to ER contribute to its success. The effects of ER proved to be positive across all included language domains (reading comprehension, vocabulary, decoding/fluency, motivation, writing, oral proficiency, general language proficiency) and target populations, with effect sizes ranging from small to medium. Variability in effect sizes was affected by two intervention characteristics: when learners' text choice was limited and when some form of accountability was included, effects were larger than when these elements were absent. Additionally, effect sizes were larger when the members of a sample had different first languages. This meta-analysis supports the assumption that encouraging second/foreign language learners to engage in sustained, independent, and self-selected reading promotes a range of second/foreign language skills. Since ER seems to be underutilized as an educational approach in second/foreign language teaching, our outcomes can be taken as a stimulus to promote its use. At the same time, further research is needed, both in the context of second language learning and in younger populations.

Keywords Extensive reading · Second language learning · Foreign language learning · Meta-analysis

Introduction

Extensive reading (ER) has emerged as a compelling strategy for enhancing learners' language proficiency, both in the realm of second language learning and foreign language learning. In second language learning, learners acquire an additional language that is commonly spoken in the country where it is learned. In contrast, foreign language learners study a language that is not used as a native language in the learner's immediate environment. Unlike traditional intensive reading (IR) approaches, which focus on detailed analysis of short texts, extensive reading encourages wide reading: second and foreign language learners immerse themselves in large amounts of text. During extensive reading sessions, learners read at their own pace, individually and silently (Day & Bamford, 1998, 2002), and the focus is on reading for enjoyment. Although there is variability in the conceptualization of extensive reading (Waring & McLean, 2015), the regularly cited *top ten principles of extensive reading* published by Day and Bamford (2002) provide an overview of elements that seem to be central to the implementation of extensive reading in second and foreign language teaching practice: (1) the reading material is easy (i.e., attuned to learners' reading proficiency), (2) a variety of reading material on a wide range of topics must be available, (3) learners choose what they want to read, (4) learners read as much as possible, (5) the purpose of reading is usually related to pleasure, information and general understanding, (6) reading is its own reward, (7) reading speed is usually faster rather than slower, (8) reading is individual and silent, (9) teachers orient and guide their students, and (10) teachers are role models (i.e., teachers read themselves during ER sessions).

Extensive Reading: Theoretical Rationale and Previous Research

Extensive reading is grounded in a combination of theoretical perspectives that highlight three interrelated mechanisms by means of which effects are assumed to occur. First, extensive reading is expected to promote reading fluency by supporting *implicit learning* (Suk, 2016): it is assumed that the development of automatized processing of text involves “the tacit learning of co-occurrence patterns” (Suk, 2016, p. 74) and that this necessitates frequent exposure to textual input (Ellis, 2008; Grabe, 2009). Frequent exposure supports automatic word recognition: it allows the strengthening and stabilization of lexical entries and networks (Nation, 2009). Second, it is assumed that language is acquired by understanding messages and, consequently, that language learners should be exposed to *comprehensible input* (Krashen, 1989), an assumption that is central to Krashen's Input Hypothesis (IH; 1981, 1982, 1992). Comparable to the notion of implicit learning, the IH posits that language learning is a matter of acquisition rather than the consequence of explicit teaching, and that acquisition occurs when input contains “structure a bit beyond our current level of competence ($i + 1$)” (Krashen, 1992, p. 21). Krashen suggests that language learners do not have to be deliberately provided with such input: he assumes that, if communication is successful and there is sufficient input, $i + 1$ will occur naturally.

The IH concerns language learning in general, but Krashen specified it to reading as well (in the so-called Reading Hypothesis, see Krashen, 1992). Although the IH has been criticized for being untestable—Krashen has not clarified how to operationalize $i+1$ —, many theories since have acknowledged the fundamental role of input in language learning (Lichtman & VanPatten, 2021). Third, it is assumed that the amount of exposure is moderated by *motivational factors triggered by choice*. In his Monitor Model (that also includes the IH), Krashen (1982) hypothesized the presence of an Affective Filter (see also Dulay & Burt, 1977): variables such as motivation can facilitate or prevent input being used for language acquisition. The notion of choice, therefore, is an important component of extensive reading: following Ryan and Deci's (2000) Self-Determination Theory, the experience of autonomy is assumed to promote intrinsic motivation. Self-selection of books thus likely lowers the Affective Filter and increases the likelihood of language input being used.

Over the years, effect studies of extensive reading have been summarized in a number of meta-analyses. A meta-analysis by Krashen (2011) of 14 pretest–posttest comparisons indicated significant growth for learners in ER programs on two measures of reading comprehension: cloze tests (Cohen's $d=0.73$) and general reading comprehension tests ($d=0.88$). Krashen additionally tested one of the assumed mechanisms behind the impact of ER, that is, whether effects were achieved by increased access to books. This hypothesis was only partially supported: the number of books available per student was associated with reading comprehension, but not with cloze test scores. A meta-analysis by Kim (2012) of 21 (quasi-)experimental studies reported positive effects of extensive reading on cognitive outcomes ($d=0.62$)—both in the domain of reading (e.g., comprehension, fluency) and outside (listening and writing)—and on affective outcomes (e.g., motivation, attitude, $d=0.40$). Kim (2012) also suggested a U-shaped relationship between effects and age, where children and adults benefit more than adolescents, but this moderator effect was not formally tested. A meta-analysis by Nakanishi (2015) of 34 pretest–posttest comparisons, including 18 experimental vs. control group studies, showed that extensive reading is effective in promoting second language learners' reading proficiency, including reading speed, reading comprehension, vocabulary, and grammar; learners in ER programs scored better on posttests compared to pretests ($d=0.71$) and outperformed learners in control groups ($d=0.46$). Nakanishi additionally tested whether intervention effects were affected by learners' age and length of instruction. His conclusions were ambivalent: he suggested that “the effect of extensive reading increases with older participants” (p. 26) and that “longer length of instruction period could improve students' proficiency in English” (p. 27), but at the same time had to conclude there were no statistically significant differences among subgroups. A meta-analysis by Jeon and Day (2016) of 51 experimental vs. control group studies and 20 pre-post design studies revealed a positive effect of extensive reading on learners' reading proficiency, including reading rate, reading comprehension, and vocabulary; learners in ER programs scored better on posttests than pretests ($d=0.79$) and outperformed learners in control groups ($d=0.57$). The authors tested the effects of several moderators and found five to yield significant effects: effect sizes increased with publication year, were larger in EFL settings, for adults, when web instead of paper texts were used, and when ER was part of a

course or offered as an extracurricular activity. In all cases, however, Jeon and Day warned that effects should be interpreted with caution, since particular studies (more recent studies, studies in EFL contexts, with adult learners, and studies where paper texts were used and where ER was part of a reading course) were heavily overrepresented. Finally, a meta-analysis by Liu and Zhang (2018) of 21 studies demonstrated an impact of extensive reading on foreign language learners' vocabulary learning; learners in ER programs scored better on posttests compared to pretests ($d=3.26$ based on $k=21$) and outperformed learners in control groups ($d=1.32$ based on $k=17$). Moderator analyses revealed that effects were larger for shorter programs, when graded readers—books designed to help language learners improve their language skills and written with simplified language and controlled vocabulary to match different proficiency levels—were used, and when participants were asked to answer comprehension questions about the books read.

Variability in the Implementation of Extensive Reading

Research on extensive reading has thus quite consistently shown its impact on a range of second and foreign language learning domains. However, some caveats must be made. In a number of meta-analyses, conclusions were fully (Krashen, 2011) or partly (Jeon & Day, 2016; Liu & Zhang, 2018; Nakanishi, 2015) based on weaker research designs (i.e., studies without a control group). Importantly, the meta-analyses only gave limited insight into which elements might be responsible for the positive effects of ER: as shown above, some evidence was found for the effects of the number of books available, program length, medium and type of reading materials, the position of ER in the curriculum, and the use of certain pedagogical additions, but some of these outcomes appeared only tentative (Jeon & Day, 2016; Krashen, 2011; Nakanishi, 2015). Moreover, these moderators only cover part of the variability in the application of ER.

A selection of studies available shows quite a number of deviations in the use of the key elements suggested by Day and Bamford (2002). With respect to the selection of reading materials (“The reading material is easy” and “A variety of reading material on a wide range of topics must be available”), some programs gave learners a free selection among all available materials, while others limited the learners' text choice. For example, Yamamoto (2011) only allowed students to select books from the Oxford Bookworms series matching their current vocabulary level, while Chen et al. (2013) allowed students to read beyond their current level if they liked the challenge. In attuning text choice to learners' reading level, some programs also promoted the use of the aforementioned graded readers (Yang et al., 2021). Another example of variability lies in how the role of teachers takes form (“teachers orient and guide their students”). In some programs, teachers assisted students in book selection (Martinez, 2017), while in other programs teachers provided learners with help during reading—for example by answering questions and by guiding learners who had difficulty recognizing words (Lituanas et al., 2001)—or wrote brief comments to learners' written responses to the books they read (Lee & Schallert, 2015).

Besides diversity in the conceptualization of key elements, programs also vary in the kinds of additions they make to the core elements of extensive reading. For instance, follow-up activities are included in some extensive reading programs, though their presence and form vary across studies. That is, some but not all programs asked learners to account for their reading, for instance in the form of book reports (Lee & Schallert, 2015), quizzes about the book's content (Milliner, 2019), and/or reading logs (Song & Sardegna, 2014). In addition, some programs let students interact with each other after reading (Yilmaz et al., 2020). Additionally, even though a key element of extensive reading is that reading should be "its own reward", programs sometimes provided (external) rewards for reading, for example by giving out course credits (Shimono, 2019). Finally, some programs did not only facilitate reading in the educational context but also actively promoted learners to read outside of the classroom (Jeon, 2008).

Possible Moderators of Extensive Reading Effects

Extensive reading is assumed to elicit the same mechanisms as approaches such as Independent Silent Reading (ISR) do (Garan & DeVoogd, 2008). It might thus be subject to the same constraints (Reutzel et al., 2010): self-selection might be difficult for less experienced readers, simply allotting time for reading does not guarantee that students read, there is a lack of accountability (students do not have to demonstrate their reading), and there is, in principle, little opportunity for interaction between readers, even though this might be beneficial for their motivation to read. Recently, Merke et al. (2024) conducted a meta-analysis to test whether additions to ISR that are aimed to overcome such barriers contribute to its effects: they examined the impact of limiting and supporting book selection, of measures to increase accountability (particularly, student-teacher conferences, tests, quizzes, and reading logs), of help/instruction by teachers, of the provision of opportunities to interact about books (between students and teachers, among students), of the use of rewards and digital support programs (e.g., to monitor students' reading), and of promoting the use of books outside the educational context (at home). Based on the outcomes of 51 studies, Merke et al. found that, overall, such additions have a small but significant effect on reading proficiency (Cohen's $d=0.27$). In the current meta-analysis, we used Merke et al.'s taxonomy to test the moderating effects of intervention characteristics.

Additionally, we examined the moderating effects of sample characteristics. Extensive reading is applied in second as well as in foreign language learning contexts. Both have characteristics that may promote ER effects. In FL contexts, for instance, students usually have relatively little language exposure outside the school domain; ER, then, is an important means to increase input. L2 learners, on the other hand, may be more motivated to increase their skills because they are dependent on them in everyday academic and societal contexts, and may thus be more willing to invest in frequent reading. Consistent with the first assumption, in their meta-analysis, Jeon and Day (2016) found ER to yield larger effects for EFL than for ESL students, but also noticed an imbalance in research attention: there were considerably

more studies on the former than the latter learners. An updated meta-analysis might thus yield more robust results. Because Merke et al. (2024) found that the aforementioned additions to ISR were particularly beneficial for students at risk of reading failure, we also aimed to test moderator effects of risk indicators such as socioeconomic status. Alternatively, at-risk status might impede the effects of ER because struggling readers are generally less prone to engage in reading (Schiefele et al., 2012; Toste et al., 2020) and might thus benefit less from ER. In a Dutch study on ISR, evidence was found for such a reverse effect. Van der Sande et al. (2019) established that for students, who were, at the start of the school year, infrequent readers (as assessed by a title recognition test), ISR was counterproductive: the more time schools made available for ISR, the less these students tended to read, most likely because they were confronted with an activity they felt resistant to. As described earlier, Jeon and Day (2016) found that age moderated the effects of ER on reading proficiency: adult readers seemed to benefit most. The authors suggested that adults' larger reading experience, background, and vocabulary knowledge make them better equipped for reading extensively than children and adolescents are. However, because other meta-analyses were ambivalent in their conclusions about age effects (Kim, 2012; Nakanishi, 2015), we decided to include age/grade as a possible moderator as well.

The Current Study

In this meta-analysis, we aimed to gain more insight into the characteristics of extensive reading that contribute to its success. Our study was built up around two research questions:

1. What are the effects of extensive reading on the development of learners' second and foreign language skills?

This question allowed us to update the conclusions of previous meta-analyses. We defined both "learners" and "language skills" broadly: we targeted child, adolescent, and adult learners, and we included multiple language domains: decoding and fluency, grammar, reading comprehension, reading motivation, speaking, vocabulary, and writing.

2. To what extent are these effects moderated by (a) sample, (b) intervention, (c) study, and (d) measurement characteristics?

Sample moderators (a) are learner type (i.e., second vs. foreign language learners), first language of participants (same vs. different), and educational level (primary, secondary, tertiary education, other). Intervention moderators (b) are interventionist (was the intervention executed by a researcher?), whether or not the intervention included text limitation, help with text selection, accountability, help/instruction before or during ER, interaction, rewards, a digital support program,

home activities, the provision of books at home, graded readers, the type of text medium, and the duration of the intervention in weeks. Study moderators (c) are peer review, design (quasi-experimental vs. experimental), random assignment, randomization level, and study quality. We will discuss elaborate definitions of these moderators in the Method section.

Method

Search Strategy and Study Selection

We conducted a search in five electronic databases: ERIC, PsycINFO, Web of Science Core Collection, Scopus, and Google Scholar. We formulated the search strategies together with an experienced information specialist. Two sets of keywords were combined: keywords referring to extensive reading and similar approaches (e.g., independent silent reading, sustained silent reading) and keywords reflecting the language learning context (e.g., second language learning, foreign language learning, bilingual education; see Appendix 1 for a full overview of the search syntax per database).

In addition to peer-reviewed articles, we included other publications, such as dissertations and book chapters, to generate a full overview of the available information and to examine publication bias. We included studies that were published in English and met the following criteria: (a) the study encompassed an intervention that was aimed at enhancing second and/or foreign language learning by means of extensive reading, (b) the study contained an experimental group and a control group, and (c) the study provided effect sizes or information allowing the calculation of effect sizes (sample size, means, and SD's, or results of statistical testing). There were no limitations concerning publication date. Studies that focused on first language learning and/or targeted specific groups (e.g., learners with learning, emotional, or behavioral disorders) were excluded.

In the initial search, which was carried out on 26 January 2023, 1710 titles were identified. The first 500 titles were screened on title and abstract according to the inclusion and exclusion criteria by Author 1, 2, 4, and 5; the other 1210 titles were screened by Author 1 and 2. Full texts of potentially relevant studies were assessed on the same inclusion and exclusion criteria to compile a final selection. If articles were not directly accessible, we tried to retrieve them by contacting the authors. For 44 potentially relevant articles, we were not able to retrieve the full text. If studies were eligible, but the statistical data reported were insufficient to calculate the required effect sizes for the meta-analysis, we e-mailed the corresponding author to request the necessary information. In this way, we were able to collect additional data for two studies. In addition, we consulted the reference lists of previous meta-analyses, and the reference lists of studies that were selected based on our database search. This led to the inclusion of six additional studies. Finally, after combining duplicate studies, our initial dataset consisted of 68 studies. The update of our initial search, which was carried out on 14 April 2025 in the same five databases, led to the inclusion of

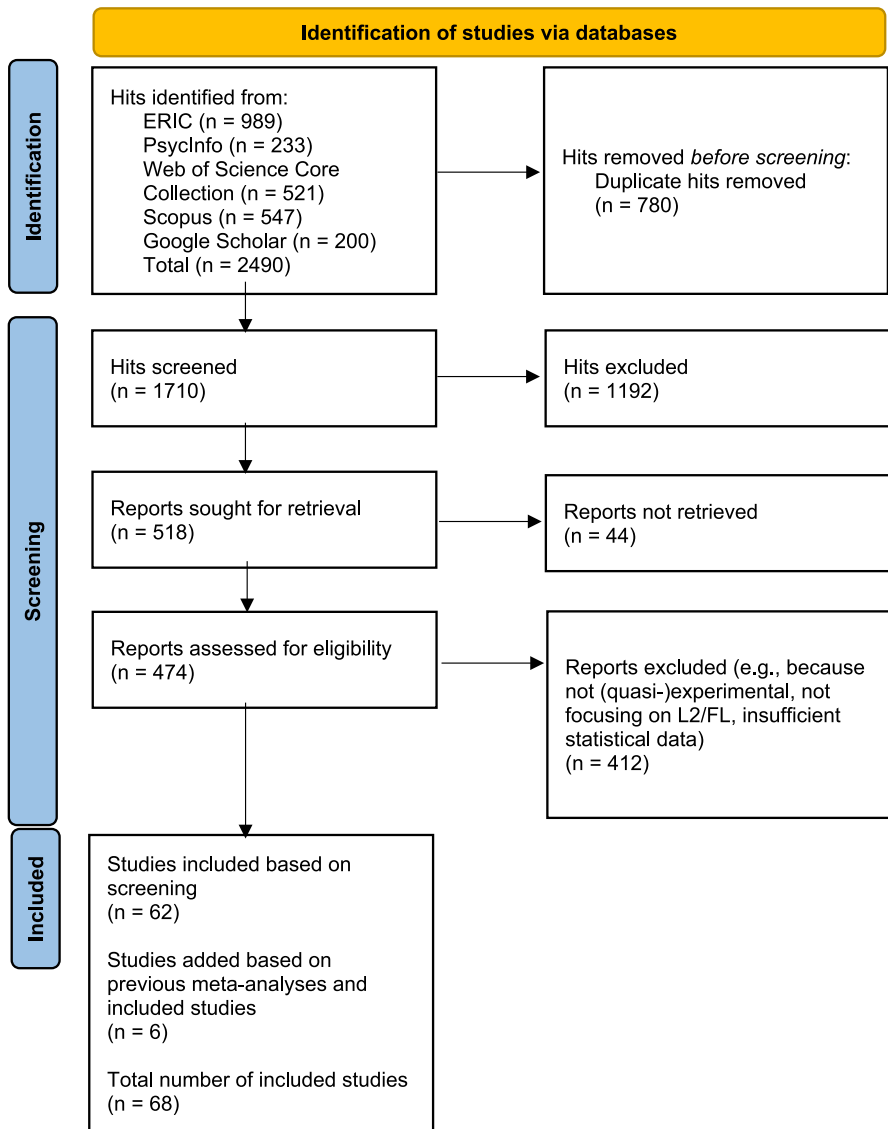


Fig. 1 PRISMA flow diagram for the initial search

six additional studies (following the same procedure as during the initial search). One of these studies (published in a peer-reviewed journal) was combined with a study already included in the initial search (published as a PhD manuscript), which contained the same data on part of the variables. Figures 1 and 2 show the PRISMA flow diagrams of the initial and updated search, respectively.

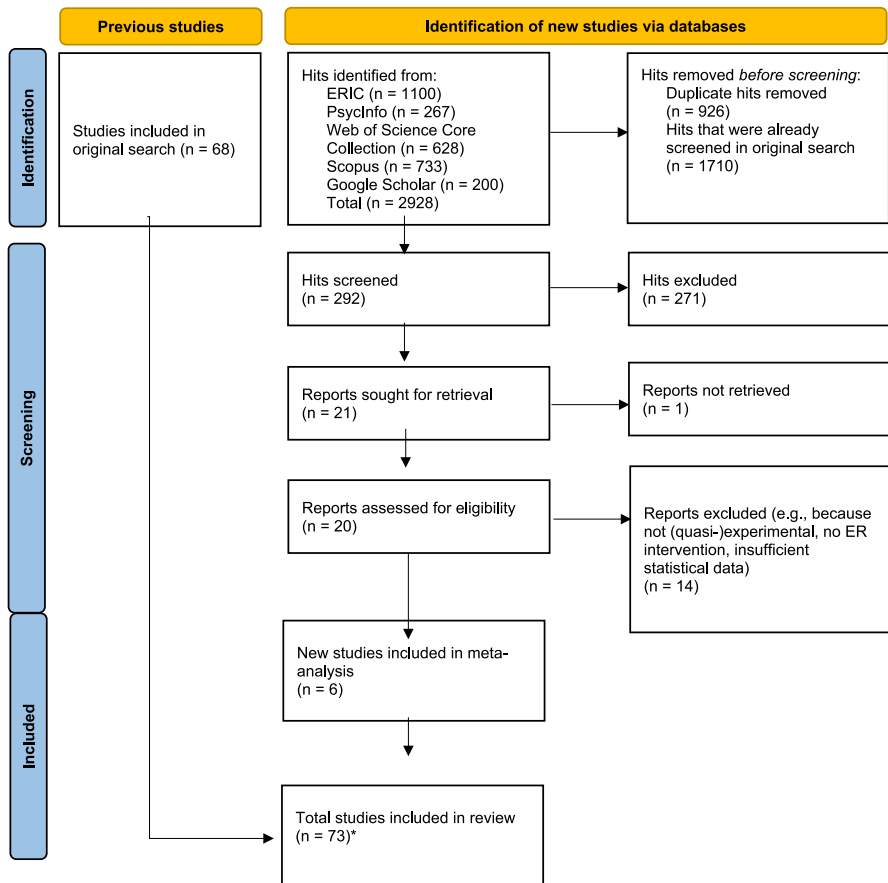


Fig. 2 PRISMA flow diagram of the updated search. Asterisk (*) indicates that one of the new studies (published in a peer-reviewed journal) was combined with a study included in the original search (published as a PhD manuscript) with the same data on part of the variables

Coding Scheme

All studies were double coded by Author 1 and 2 (initial search) and by Author 1 and 5 (updated search) according to a coding scheme with the following sections: bibliographic information, sample characteristics, intervention characteristics, study characteristics, and measurement characteristics.

Bibliographic Information

We registered the title of the article, the author name(s), and the publication year.

Sample Characteristics

We registered whether the intervention focused on second and/or foreign language learners (and which language was to-be-learned). Students were considered second language learners when they were learning an additional language that was commonly spoken in the country they live in (e.g., a student who speaks Spanish at home but learns English at school in the USA is a second language learner of English); students were considered foreign language learners when the language they were learning was not commonly spoken in their country but is typically learned in a classroom setting (e.g., a student who speaks Spanish at home and learns English at school in Spain is a foreign language learner of English). We also coded whether the students in the sample had the same or different first language(s) (and which language(s) this concerned), and what level of education they were in (primary, secondary, tertiary education, or other). We also specified the mean age of the students, the percentage of women in the sample, the percentage of students with low socioeconomic status, and the percentage of students with an immigrant status. However, as the studies did not provide enough information on these latter four variables, we excluded them from the analyses.

Intervention Characteristics

We coded the name of the country in which the study took place, the name of the intervention, the duration of the intervention (in weeks and hours, number of sessions, length of each session in minutes, and total reading time at home), and whether it was executed by the researcher (or e.g., a teacher). As little information was provided on several duration variables, we only included the duration of the intervention in weeks in our analyses. Regarding the content of the intervention, we coded (a) whether students were limited in their text choice (e.g., based on their reading level) or whether they were allowed boundless choice (“text limitation”), (b) whether students were supported in their text choice (“text selection”, e.g., by teacher recommendations based on their reading level or interests), (c) whether students were held accountable for their reading (“accountability”, e.g., by having to keep a log about what they were reading or take a short test about their book), (d) whether students were supported before or during their reading (“help/instruction”, e.g., by student–teacher conferences to resolve comprehension problems or strategy lessons), (e) whether opportunities for interaction were created (“interaction”, e.g., student–student talks or class discussions about the books students were reading), (f) whether students were given rewards for reading (“reward”, e.g., obtaining credits), (g) whether the intervention included technological facilities (“digital support program”, e.g., digital programs for quizzes), (h) whether the intervention actively promoted the home use of books for extensive reading (“home activities”), and (i) whether students were allowed to take books home or use them outside of the classroom (“home books”). In case a study did not provide information on a characteristic, we assumed that it was not part of the intervention (thus, a “no” was coded). Concerning text characteristics, we coded the genre of the texts/books provided (“genre”: narrative, expository, both), whether the texts/books involved

graded readers (“graded readers”), the number of texts/books that needed to be read by the students during the intervention, and the text medium (“text medium”: print, digital). As there was not enough information on text genre and the number of texts/books to-be-read, these variables were left out of the analyses.

Study Characteristics

We checked whether the study was published as a peer-reviewed article. In addition, we recorded information on the design of the study, distinguishing between experiments and quasi-experiments. Studies were only coded as an experiment if randomization was applied at the individual level. If classes or schools were randomly assigned to the experimental or control condition, the design was considered quasi-experimental. For students in the control groups, we coded whether they were exposed to a business-as-usual approach during the period of intervention or whether they participated in another intervention. However, we had to exclude this variable from the analyses, as it was not always clear what the business-as-usual approach entailed. Also, in some studies the intensive reading approach was described as “business-as-usual”, while in other studies it was posed as “another intervention”. Finally, we assessed the quality of the included studies, based on guidelines by Austin et al. (2019) and following the quality assessment of Merke et al. (2024). We made an index (minimum=0; maximum=6) on the basis of three criteria (design, implementation fidelity, statistical analyses), with per criterion three possible scores: 2 points (exemplary), 1 point (acceptable), and 0 points (unacceptable) (see Appendix 2 for an overview of the criteria).

Measurement Characteristics

For each effect measure, we coded which second/foreign language learning domain it involved: decoding and fluency, grammar, reading comprehension, reading motivation, speaking, vocabulary, and writing. Additionally, we incorporated a category “general language proficiency”. This category included the outcomes of tests that provided an overall score by determining the combined proficiency for two or more domains, such as the TOEIC test for listening and reading (e.g., Goto, 2021; Rezaee et al., 2021). Further, we registered which type of instrument was used (test, self-report, observation, teacher report), whether this instrument was an (adapted) existing instrument or an instrument developed for the current intervention, whether the posttest was executed immediately after the intervention or whether it was delayed (i.e., a follow-up measurement; and if so, how many weeks), and whether the same instrument was used for the pre- and posttest. As all language learning domains were measured by means of tests, except for motivation—which was measured by means of self-reports—the variable “type of instrument” was not informative enough to be included in the analyses. In addition, it was often unclear whether the instrument used was an (adapted) existing instrument or not, and whether this instrument was the same for pre- and posttest. As such, we also excluded these two variables from the analyses. Finally, we entered the statistical information necessary to compute effect sizes—mean, *SD*, *n*, test statistics such as *t* or *F*—and, if provided, the effect sizes (Cohen’s *d*, Hedges’ *g*, or η^2) as reported by the authors.

Coding Procedure

The coding scheme was developed by Author 1 and 2. A first version of the coding scheme was tried out on a sample of three studies by Author 1, 2, and 5. These authors coded the studies, compared their codes, and adjusted the scheme based on their comparisons. Subsequently, Author 1 and 2 coded the remaining studies; thus, all studies were double-coded. Interrater agreement was 87% (range: 77–99%, Cohen's $\kappa=.479-.871$). For the variable “help/instruction”—i.e., assessing whether students were supported before or during their reading—the interrater agreement was relatively low (77%, $\kappa=.479$). This was mainly caused by a lack of clarity in the information provided by primary studies. Discrepancies were discussed until consensus was reached. In case of uncertainties, Author 5 was consulted for a final decision. Author 1 and 5 double-coded the six additional studies that were included based on the update of the search.

Data-Analysis

As some of the 73 selected studies included more than one experiment, experimental condition, or subsample, “experimental comparison” was used as the basis for the analyses. The total number of experimental comparisons was 86 (including follow-up comparisons). For the analysis of overall effects, we computed a weighted effect size (Cohen's d) for each experimental comparison, for which we used the available statistical information. When multiple effect measures were used to assess a certain outcome within an experimental comparison (e.g., multiple measures of reading motivation), we calculated the mean of the effect sizes for these measures (i.e., leading to one overall measure for reading motivation). When studies included both an overall measure and several subscales, we selected the overall scale.

If possible, we used both pre- and posttest data for the calculation of effect sizes. Some studies reported gain scores instead of pre- and posttest means and standard deviations. If means and standard deviations and gains were not reported, we calculated the effect sizes based on other available statistical information, such as t -values, F -values, and p -values, together with information on the sample size. If this other statistical information was insufficient, we used the effect sizes provided by the primary authors.

All analyses were carried out by Author 1 and 5 using Comprehensive Meta-Analysis statistical software (version 4.0). The analyses were based on the random effects model, taking the heterogeneity across studies into account. To account for differences in sampling error related to sample size, random effects weigh the mean effect size by the variance of the sample and the variance between studies. To explain between-study variance, we conducted moderator analyses, based on subgroup analysis for categorical variables and meta-regression analyses for continuous variables. To interpret the effect sizes, we used Cohen's d (1988), distinguishing between small ($d>.20$), medium ($d>.50$), and large ($d>.80$) effects.

Publication Bias

The risk of publication bias appears to be limited. First, the nonsignificant moderator effect of the variable “peer review” (see Table 4 in Results) suggests the absence of publication bias (Borenstein et al., 2021). In addition, we generated a funnel plot based on the overall effect sizes for all second and foreign language learning skills (Fig. 3). Figure 3 shows generally small effect sizes, that are quite evenly distributed around the mean. Figure 3 also reveals five outliers. However, exclusion of these outliers hardly affected the overall intervention effects of the meta-analysis (Cohen’s $d=0.39$; $SE=0.07$). Finally, we conducted Egger’s regression test (Egger et al., 1997), which revealed a non-significant intercept ($\beta = -0.50$, $p=.466$), additionally suggesting a lack of publication bias.

Results

Descriptive Information

The 73 studies included in this meta-analysis comprise 82 interventions (for an overview of the interventions and their sample, intervention, and study characteristics, see Appendix 3). Most studies were peer reviewed ($k=67$, 92%). The average score on the study quality index was 2.93 ($SD=1.17$; ranging from 0 to 5), which is in the middle of the 0 to 6 index. This relatively low score was mostly due to a low score on implementation fidelity ($M=0.38$, $SD=0.49$); 45 out of 73 studies did not report implementation fidelity. No study was excluded on the basis of study quality score. Out of the 82 interventions, all but two (which involved learners of Spanish) focused on English language learning ($k=80$, 98%). Most of the interventions took place

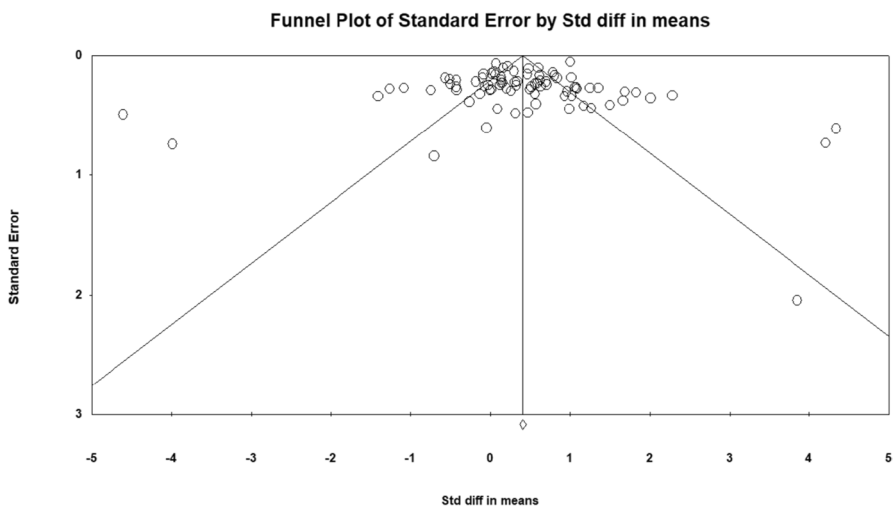


Fig. 3 Funnel plot of the overall effect sizes for all second and foreign language learning skills

in a foreign language learning context, meaning that the language that was to-be-learned by the students was not an official language of the country the intervention was conducted in ($k=78$, 95%). The studies were mostly located in Asia ($k=44$, 60%). Only four studies were carried out in a second language learning context, namely that of English learning in the USA. We excluded this variable from further moderator analyses, as the number of studies ($k=4$) did not reach the threshold of five (Kontopantelis & Reeves, 2010). The studies were generally quasi-experimental ($k=56$, 77%), and the comparisons focused on university students ($k=46$, 56%); other educational levels were less well represented (primary education: $k=7$, 9%; secondary education: $k=19$, 23%; other: $k=9$, 11%; missing: $k=1$, 1%). Although many studies did not provide conclusive information on gender, the percentage of females in the interventions ranged from 0% (all male) to 100% (all female). The duration of the interventions varied from 4 weeks to a full academic year.

Regarding the intervention characteristics, 37 of the interventions were carried out by the researcher (45%) compared to 31 teacher-led interventions (38%) (missing: $k=14$, 17%). Most interventions did not limit students' text choice ($k=57$, 70%), did not provide students with help during text selection ($k=73$, 89%) nor gave additional instructions before and/or during reading ($k=52$, 63%). In most interventions, students had to account for their reading ($k=59$, 72%). Only a limited number of interventions allowed for interaction after reading ($k=26$, 32%) or gave students a reward for their reading ($k=21$, 26%), for instance in the form of course credits. In 11 interventions (13%), a digital program was used to support students' reading, often in the form of digital quizzes, such as the MoodleReader. In about half of the interventions, students were actively stimulated to also read texts outside of the classroom ($k=44$, 54%). In 60 percent of the interventions, students were allowed to take books home ($k=49$). Graded readers were quite frequently used ($k=58$, 71%), and most interventions made use of printed texts ($k=74$, 90%).

Inferential Statistics

Intervention Effects

We first analyzed the overall intervention effects based on 86 experimental comparisons. This revealed a small, significant positive effect (Cohen's $d=0.41$; $SE=0.07$). As there were only a limited number of follow-up comparisons ($k=4$), we decided to only include immediate effects in our analyses ($k=82$; Cohen's $d=0.38$, $SE=0.07$). In addition, there were only a limited number of second language learning studies ($k=4$). We conducted a sensitivity analysis without these L2 studies to check whether this affected the intervention effects and moderator analyses. As this was not the case (intervention effects: $k=82$, Cohen's $d=0.40$, $SE=0.08$), we decided to keep the L2 studies in our analyses.

Subsequently, we analyzed the effect sizes per effect measure: decoding and fluency, grammar, reading comprehension, reading motivation, speaking, vocabulary, writing, and general language proficiency. As listening, speaking and grammar

Table 1 Effect sizes per effect measure

Effect measure	<i>k</i>	<i>d</i>	<i>SE</i>	Variance	95% CI	<i>z</i>	<i>Q</i>	<i>I</i> ²
Reading comprehension	52	0.31	0.08	0.006	[0.16–0.47]	3.90***	433.02***	88.22
Vocabulary	31	0.50	0.14	0.018	[0.24–0.77]	3.72***	315.06***	90.48
Decoding and fluency	23	0.38	0.14	0.021	[0.10–0.66]	2.63**	160.33***	86.28
Motivation	17	0.34	0.17	0.030	[0.004–0.68]	1.98*	161.57***	90.10
Writing	14	0.31	0.13	0.016	[0.07–0.56]	2.49*	79.42***	83.63
Oral proficiency	11	0.42	0.11	0.012	[0.21–0.64]	3.84***	47.49***	78.94
General language proficiency	9	0.53	0.16	0.025	[0.22–0.84]	3.37***	56.23***	85.77

* $p < .05$; ** $p < .01$; *** $p < .001$

Table 2 Effect sizes for sample characteristics

Moderator	Category	<i>k</i>	<i>d</i>	<i>SE</i>	95% CI	<i>Q</i>
First language	Same	77	0.33***	0.08	[0.18–0.48]	6.19*
	Different	5	1.03***	0.27	[0.50–1.55]	
Educational level	Primary education	7	0.32	0.34	[-0.13–0.77]	1.52
	Secondary education	19	0.53***	0.14	[0.25–0.80]	
	Tertiary education	46	0.36***	0.09	[0.18–0.55]	
	Other	9	0.04	0.56	[-1.05–1.13]	

were not often included as effect measures in the interventions, we decided to combine them into one effect measure: oral proficiency. The interventions had small to medium, significant positive effects on all effect measures, as shown in Table 1.

Moderator Analyses

The significant *Q*-statistics in Tabel 1 show that there was variability in effect sizes for all outcomes. To explain this variability in effect sizes, we conducted moderator analyses based on sample, intervention, and study characteristics. The moderator analyses were conducted over all effect measures combined. This approach was chosen to increase statistical power and reduce the risk of Type I errors associated with conducting multiple separate analyses. By combining effect measures, the analysis provides a more comprehensive understanding of whether moderators exert a general influence on intervention effectiveness, rather than being limited to specific outcome domains. The outcomes of the moderator analyses are shown in Tables 2, 3, and 4.

Sample Characteristics First language was a significant moderator of intervention effects: interventions in which students had mixed language backgrounds showed larger effects. There was no significant effect for educational level.

Table 3 Effect sizes for intervention characteristics

Moderator	Category	<i>k</i>	<i>d</i>	<i>SE</i>	95% CI	<i>Q</i>
Researcher	Yes	37	0.22	0.13	[-0.03–0.47]	1.11
	No	31	0.40***	0.11	[0.18–0.62]	
Text limitation	Yes	25	0.73***	0.13	[0.48–0.99]	10.52**
	No	57	0.22*	0.09	[0.04–0.40]	
Text selection	Yes	9	0.46	0.26	[-0.04–0.96]	0.13
	No	73	0.37***	0.08	[0.21–0.52]	
Accountability	Yes	59	0.51***	0.09	[0.35–0.68]	8.72**
	No	23	0.01	0.15	[-0.28–0.30]	
Help/instruction	Yes	30	0.32*	0.15	[0.03–0.61]	0.23
	No	52	0.40***	0.09	[0.23–0.57]	
Interaction	Yes	26	0.41*	0.16	[0.09–0.73]	0.05
	No	56	0.36***	0.09	[0.20–0.73]	
Reward	Yes	21	0.37**	0.13	[0.13–0.62]	0.004
	No	61	0.38***	0.09	[0.20–0.56]	
Digital support program	Yes	11	0.37	0.23	[-0.08–0.82]	0.001
	No	71	0.38***	0.08	[0.22–0.53]	
Home activities	Yes	44	0.40***	0.10	[0.21–0.59]	0.10
	No	38	0.35**	0.12	[0.12–0.59]	
Home books	Yes	49	0.37***	0.09	[0.19–0.55]	0.03
	No	33	0.40**	0.13	[0.14–0.65]	
Graded readers	Yes	58	0.47***	0.09	[0.30–0.64]	0.000
	No	11	0.47*	0.21	[0.06–0.87]	
Text medium	Digital	8	0.64*	0.27	[0.11–1.17]	1.10
	Print	74	0.34***	0.07	[0.20–0.49]	

* $p < .05$; ** $p < .01$; *** $p < .001$

Table 4 Effect sizes for study characteristics

Moderator	Category	<i>k</i>	<i>d</i>	<i>SE</i>	95% CI	<i>Q</i>
Peer review	Yes	76	0.35***	0.08	[0.20–0.50]	1.88
	No	6	0.84*	0.35	[0.15–1.54]	
Design	Experimental	15	0.58	0.34	[-0.08–1.25]	0.51
	Quasi	63	0.34***	0.08	[0.19–0.49]	
Random assignment	Yes	31	0.47**	0.16	[0.16–0.77]	0.15
	No	19	0.55***	0.13	[0.30–0.79]	
Randomization level	Individual	16	0.58*	0.30	[0.005–1.16]	0.97
	Class	58	0.28**	0.01	[0.12–0.44]	
	School	2	0.31	0.23	[-0.14–0.76]	

* $p < .05$; ** $p < .01$; *** $p < .001$

Intervention Characteristics Of the categorical intervention characteristics, only text limitation and accountability were significant moderators. Interventions in which students were somehow limited in their text choice (e.g., based on their reading level) had larger effects, as did interventions in which students were held accountable for their reading. The continuous variable that indicated the duration of the interventions in weeks was not found to be significant ($\beta = -0.002$, $SE = 0.004$, $p = .669$).

Study Characteristics We examined the effects of four study characteristics: peer review, design, random assignment, and randomization level. The moderator analyses did not reveal any significant effect of these variables. In addition, a meta-regression analysis did not reveal a moderator effect of study quality ($\beta = 0.06$, $SE = 0.06$, $p = .346$).

Discussion

Summary of Findings

The aim of this meta-analysis was to assess the impact of extensive reading on second and foreign language learning and, more specifically, to examine which elements contribute to its success. The short-term effects of extensive reading proved to be positive across all included language domains and target populations, with effect sizes ranging from small to medium. No long-term effects could be established due to a lack of follow-up measurements. Variability in effect sizes was affected by two intervention characteristics: when students' choice of texts was limited and when some form of accountability was included, effects were generally larger than when these elements were absent. Additionally, effect sizes were affected by the question whether the members of a sample had the same or different first languages: effects were larger in the latter case.

Interpretation of Findings

The overall positive effects we found support the assumed benefits of extensive reading for second and foreign language learning (Ng et al., 2019), and corroborate the outcomes of earlier meta-analyses (Jeon & Day, 2016; Kim, 2012; Krashen, 2011; Liu & Zhang, 2018; Nakanishi, 2015). Apparently, encouraging learners to engage in individual, silent reading of longer target language texts of their own choice, but attuned to their proficiency level promotes a variety of second and foreign language skills. Interestingly, this finding differs from the results of sustained self-selected reading in other contexts. Erbeli and Rice (2022), for instance, reviewed 14 experimental and quasi-experimental studies on the effects of Independent Silent Reading (ISR) but found no support for the presence of its effects on reading outcomes. Bus et al.'s (2024) meta-analysis of 47 (quasi-)experiments revealed only a marginal overall effect of ISR (Hedge's $g = 0.08$); specific effects varied from small in the case of word reading skills (Hedges' $g = 0.21$) to negative in the case of reading comprehension (Hedges' $g = -0.14$).

One explanation for the difference in outcomes between these studies might be that target populations differ in initial reading motivation (Briggs & Walter, 2016): while our meta-analysis mostly included university students learning English as a foreign language, the reviews by Erbeli and Rice (2022) and Bus et al. (2024) targeted younger students (K-12) learning their native language. Possibly, the adult EFL learning context makes the relevance of engaging in reading for language learning more salient: since students' proficiency in English has more direct consequences for achieving their educational goals, students may be more inclined to respond positively to interventions that promote their language development. Research seems to support this hypothesis. Kim (2011), for instance, examined the components of EFL reading motivation among Korean college students. Kim distinguished four motivational dimensions—goal-oriented motivation, utility value of L2 reading, intrinsic motivation, and avoidance of reading—and found that students' motivation for reading in English was mainly guided by the former two. Conversely, for native language learning, Van der Sande et al. (2019) found that increased time allocated to ISR was counterproductive for Dutch primary education students who were infrequent readers: the more time given to ISR, the less these students read. This was most likely the case because these students were confronted with an activity they felt resistant to. This aversion to reading might be less prominent in second and foreign language learners.

The outcomes of our analysis of intervention characteristics highlight the importance of one key element of extensive reading—the selection of texts attuned to students' reading level—and of the addition of accountability (through the use of reading logs or other activities). These outcomes align with the results of a recent meta-analysis by Merke et al. (2024). These authors tested the effects of interventions that meet four important limitations of ISR (Reutzel et al., 2010), which might also hold for extensive reading: (a) ISR capitalizes on self-selection of books, but some students find it difficult to find books that match their reading level and interests; (b) simply allotting time for reading does not guarantee that students read; (c) ISR lacks accountability, while some students may need an external stimulus to engage in reading; and (d) ISR provides only little opportunity for students to interact about what they read, while sharing reading experiences may be highly motivating. Merke et al. found positive effects on overall reading proficiency of a combination of interventions, including those found to be significant moderators in our meta-analysis: limiting text choice and accountability. In conclusion: extensive reading seems particularly beneficial when book selection is made easier and when there is an external impetus for reading.

An obvious explanation for the effect of sample composition is that the need to communicate in the second/foreign language is larger when the sample is heterogeneous: when members of a group vary in their first language, the second/foreign language is the *lingua franca*; this situation in itself can support language learning. It can be speculated that extensive reading incites increased use of the target language: students may be motivated to discuss the books they read. It must be added, though, that if such discussions occurred, they were likely initiated by students themselves: as indicated, only few interventions allowed for interaction after reading. The outcome might be an impetus for including opportunities for interaction during language classes.

Limitations and Directions for Future Research

One limitation of our meta-analysis—but also of previous meta-analyses of extensive reading studies (Jeon & Day, 2016; Kim, 2012; Krashen, 2011; Liu & Zhang, 2018; Nakanishi, 2015)—is that the underlying research is focused on specific populations: foreign language learners and university students were overrepresented. One important recommendation is therefore to conduct more research in other target groups. Originally grounded in theories of second language learning—particularly, in Krashen's (1981) Input Hypothesis—extensive reading is assumed to be a viable way of maximizing comprehensible input for second language learners. With only four out of 82 interventions assessing the effects of extensive reading in an L2 context, additional studies are certainly warranted. The same holds for age. There appears to be disagreement on the question whether larger effects of extensive reading can be expected for younger or older learners: while Nakanishi (2015) and Ng et al. (2019) suggest that learners have a much better chance of taking advantage of the benefits of extensive reading if they start early, Jeon and Day (2016) assume that adults are cognitively better equipped to benefit from extensive reading. To test the validity of either hypothesis, more studies on the application of extensive reading in primary and secondary education are necessary, as the current meta-analysis included only seven interventions for primary education and 19 for secondary education (compared to 46 for tertiary education).

An additional limitation of the current research base is the dearth of experiments examining long-term effects: only three studies (including four experimental comparisons) in our sample included tests of retention effects. Although this drawback is characteristic of much reading promotion research (Merke et al., 2024; Van der Sande et al., 2023), the question whether long-term effects occur is highly relevant. Increasing students' reading volume can be assumed to incite a process of reciprocal causation (Mol & Bus, 2011; Stanovich, 1986): more frequent reading allows students more reading practice, more practice supports reading proficiency, increased proficiency results in more self-confidence in reading and, consequently, in more motivation to read. To ascertain whether students enter such a virtuous cycle, follow-up measurements are needed (see also Nakanishi, 2015).

Other limitations are that the studies included in our meta-analysis were geographically biased (most were conducted in Asian countries) and mainly targeted English: experiments conducted in other (language) contexts are needed to increase the generalizability of findings. Moreover, cultural differences may influence students' engagement with extensive reading, suggesting a need for cross-cultural research. Finally, studies were not always complete in the information they provided about sample, intervention, study, and measurement characteristics, and in the statistical information included. This resulted in the exclusion of studies and in missing information in the analyses. We therefore call on researchers to be more precise in their descriptions.

For instance, in many studies, there was a lack of precise descriptions on the actual time on task, the time that learners spent on extensive reading during the intervention. For each study, we set out to code the duration of the intervention in terms of weeks and hours, number of sessions, length of sessions, and total hours

spent on reading at home. Unfortunately, there was only enough information available to conduct an analysis on intervention duration based on the number of weeks. This is a limitation: longer interventions may be less intense, because they offer less ER time per week, whereas shorter interventions may be more intense depending on the weekly ER time. This might also explain why no significant effect was observed for the moderator “time duration”. Due to the absence of precise data on the actual time learners spent on extensive reading, our findings regarding time as a moderator should be interpreted with caution.

Conclusion and Implications

This meta-analysis supports the assumption that encouraging second/foreign language learners to engage in sustained, independent, and self-selected reading promotes a range of second/foreign language skills. This is particularly true when ingredients are included that allow a suitable match of reader and text, and that increase accountability. Since extensive reading seems to be underutilized as an educational approach in second/foreign language teaching (Ewert, 2017; Ng et al., 2019), our outcomes can be taken as a stimulus to promote its use. At the same time, further research is needed, both in the context of second language learning and in younger populations.

Appendix 1 Search Syntax per Database

Eric

(Silent Reading/OR Sustained Silent Reading/OR Independent Reading/OR Recreational Reading/OR Individualized Reading/OR (((silent OR Independent OR Self-selected OR Extensive OR Leisure OR Recreational OR Free OR Superquiet OR Voluntary OR Pleasure OR Quiet OR Uninterrupted OR Individualized OR Individualised) ADJ3 reading) OR Drop-Everything-and-Read OR Exposure-to-Text OR Book-Flood OR Accelerated-Reader OR Wonderful-Exciting-Books OR (Read ADJ3 Relax ADJ3 Reflect ADJ3 Respond ADJ3 Rap)).ab,ti.) AND (Second Languages/OR Second Language Learning/OR Second Language Programs/OR Second Language Instruction/OR Foreign Language Books/OR “English (Second Language)”/OR Bilingual Education/OR Bilingual Students/OR (((second OR 2nd OR 2-nd OR Non-native OR Non-ative OR foreign*) ADJ3 (language* OR vocabulary*) OR ((language OR vocabulary*) ADJ3 learn*) OR l2 OR l-2 OR esl OR efl OR Bilingual* OR ell OR sll OR fl).ab,ti).

PscylINFO

(Silent Reading/OR (((silent OR Independent OR Self-selected OR Extensive OR Leisure OR Recreational OR Free OR Superquiet OR Voluntary OR Pleasure

OR Quiet OR Uninterrupted OR Individualized OR Individualised) ADJ3 reading) OR Drop-Everything-and-Read OR Exposure-to-Text OR Book-Flood OR Accelerated-Reader OR Wonderful-Exciting-Books OR (Read ADJ3 Relax ADJ3 Reflect ADJ3 Respond ADJ3 Rap)).ab,ti.) AND (Foreign Language Education/OR Foreign Language Learning/OR Foreign Languages/OR English as Second Language/OR Bilingualism/OR (((second OR 2nd OR 2-nd OR Non-native OR Non-ative OR foreign*) ADJ3 (language* OR vocabular*)) OR ((language OR vocabular*) ADJ3 learn*) OR l2 OR l-2 OR esl OR efl OR Bilingual* OR ell OR sll OR fl).ab,ti).

Web of science

TS=((((silent OR Independent OR Self-selected OR Extensive OR Leisure OR Recreational OR Free OR Superquiet OR Voluntary OR Pleasure OR Quiet OR Uninterrupted OR Individualized OR Individualised) NEAR/2 reading) OR Drop-Everything-and-Read OR Exposure-to-Text OR Book-Flood OR Accelerated-Reader OR Wonderful-Exciting-Books OR (Read NEAR/2 Relax NEAR/2 Reflect NEAR/2 Respond NEAR/2 Rap)):) AND (((second OR 2nd OR 2-nd OR Non-native OR Non-ative OR foreign*) NEAR/2 (language* OR vocabular*)) OR ((language OR vocabular*) NEAR/2 learn*) OR l2 OR l-2 OR esl OR efl OR Bilingual* OR ell OR sll OR fl))).

Scopus

TITLE-ABS-KEY((((silent OR Independent OR Self-selected OR Extensive OR Leisure OR Recreational OR Free OR Superquiet OR Voluntary OR Pleasure OR Quiet OR Uninterrupted OR Individualized OR Individualised) W/2 reading) OR Drop-Everything-and-Read OR Exposure-to-Text OR Book-Flood OR Accelerated-Reader OR Wonderful-Exciting-Books OR (Read W/2 Relax W/2 Reflect W/2 Respond W/2 Rap)):) AND (((second OR 2nd OR 2-nd OR Non-native OR Non-ative OR foreign*) W/2 (language* OR vocabular*)) OR ((language OR vocabular*) W/2 learn*) OR l2 OR l-2 OR esl OR efl OR Bilingual* OR ell OR sll OR fl))).

Google Scholar

“silent|Independent|Extensive|Leisure|Recreational|Free|Superquiet|Voluntary|Pleasure|Quiet|Uninterrupted|Individualized|Individualised|reading” “second|2nd|Nonnative|foreign|language”

Appendix 2 Criteria to Assess Study Quality

1. Design:

- 2 points (exemplary): randomized design with a sufficiently large sample (≥ 20).
- 1 point (acceptable): randomized design with an insufficient sample size (< 20) or a nonrandomized design with a sufficiently large sample (≥ 20) and evidence of pretest equivalence.
- 0 points (unacceptable): nonrandomized design with insufficient sample size (< 20) and/or no evidence of pretest equivalence.

2. Implementation fidelity:

- 2 points (exemplary): The study reports on the implementation of the intervention, for example by indicating the number of words or books participants read (relative to the intended amount), the number of quizzes completed (relative to the intended number), or how the intervention was carried out. In other words, the report addresses whether the intervention was implemented as described in the methods section. In addition, it includes an explicit statement that fidelity was adequate.
- 1 point (acceptable): The study reports on the implementation of the intervention, for example by indicating the number of words or books participants read (relative to the intended amount), the number of quizzes completed (relative to the intended number), or how the intervention was carried out. In other words, the report addresses whether the intervention was implemented as described in the methods section. There is no explicit statement that fidelity was adequate.
- 0 points (unacceptable): No information is provided regarding fidelity.

3. Statistical analyses:

- 2 points (exemplary): appropriate use of analysis matching the design of the study (e.g., ANCOVA when pre- and posttest data are available), the sample is sufficiently large (≥ 20), and effect sizes are reported.
- 1 point (acceptable): appropriate use of analysis matching the design of the study, the sample is sufficiently large (≥ 20), and effect sizes are not reported, but sufficient information is available to compute effect sizes.
- 0 points (unacceptable): inappropriate use of analysis or insufficient sample size (< 20).

A fourth criterion (Likelihood of Type I error in the case of multiple comparisons) formulated by Austin et al. (2019) was excluded, because it was often not applicable, as many studies only included one comparison.

Appendix 3 Sample, Intervention, Study, and Measurement Characteristics per Intervention

Table 5 Sample characteristics

No	Authors (year)	L2/FL	L1 language	Same L1	L2/FL language	Educational level
1	Martinez (2017)	L2	Spanish	1	English	Other
2	Yamamoto (2011)	FL	Japanese	1	English	Tertiary
3	Al-Jarf (2021)	FL	Arabic	1	English	Tertiary
4	Milliner (2019)	FL	Japanese	1	English	Tertiary
5	Tsang (1996)	FL	Cantonese	1	English	Secondary
6	Park et al. (2018)	FL	Korean	1	English	Secondary
7	Lake (2014)	FL	Japanese	1	English	Tertiary
8	Taguchi et al. (2004)	FL	Japanese	1	English	Tertiary
9	Lee and Lee (2023)	FL	Korean	1	English	Secondary
10	Huffman (2021)	FL	Japanese	1	English	Tertiary
11	Rezaee and Nourzadeh (2011)	FL	Persian	1	English	Other
12	Reynolds and Bai (2013)	FL	-	1	English	Tertiary
13	Goto (2021)	FL	Japanese	1	English	Tertiary
14	Kara (2019)	FL	Turkish	1	English	Tertiary
15	Jafarpour (2014)	FL	Persian	1	English	Secondary
16	Naghbi and Moghaddam (2013)	FL	Persian	1	English	Other
17	Beglar et al. (2012)	FL	Japanese	1	English	Tertiary
18	Beglar et al. (2012)	FL	Japanese	1	English	Tertiary
19	Fiedler Vierma (1991)	FL	English	1	Spanish	Tertiary
20	Ahour and Zarei (2016)	FL	Turkish Azari	1	English	Other
21	Vaezi and Nilforooshan (2013)	FL	Persian	1	English	Other

Table 5 (continued)

No	Authors (year)	L2/FL	L1 language	Same L1	L2/FL language	Educational level
22	Robb and Kano (2013)	FL	Japanese	1	English	Tertiary
23	McLean and Rouault (2017)	FL	Japanese	1	English	Tertiary
24	Rodrigo et al. (2004)	FL	English	1	Spanish	Tertiary
25	Park and Lee (2021)	FL	Korean	1	English	Primary
26	Park and Lee (2021)	FL	Korean	1	English	Primary
27	Burrows and Holsworth (2016)	FL	Japanese	1	English	Tertiary
28	Burrows (2012)	FL	Japanese	1	English	Tertiary
29	Lee et al. (2015)	FL	Korean	1	English	Secondary
30	Suk (2016)	FL	Korean	1	English	Tertiary
31	Yilmaz et al. (2020)	FL	Turkish	1	English	Tertiary
32	Ro and Kim (2022)	FL	Korean	1	English	Other
33	Chen et al. (2013)	FL	-	1	English	Tertiary
34	Schon et al. (1985)	L2	Spanish	1	English	Secondary
35	Song and Sardegna (2014)	FL	Korean	1	English	Secondary
36	Lee and Schallert (2015)	FL	Korean	1	English	Secondary
37	Al-Homoud and Schmitt (2009)	FL	Arabic	1	English	Tertiary
38	Mason and Krashen (1997)	FL	Japanese	1	English	Tertiary
39	Mason and Krashen (1997)	FL	Japanese	1	English	Tertiary
40	Mason and Krashen (1997)	FL	Japanese	1	English	-
41	Modirkhamene and Gowrki (2011)	FL	Kurdish	1	English	Other
42	Sheu (2003)	FL	Mandarin Chinese	1	English	Secondary
43	Sheu (2003)	FL	Mandarin Chinese	1	English	Secondary
44	Fernandez de Morgado (2009)	FL	Spanish	1	English	Tertiary
45	Iwata (2020)	FL	Japanese	1	English	Tertiary
46	Shih (2015)	FL	-	1	English	Tertiary

Table 5 (continued)

No	Authors (year)	L2/FL	L1 language	Same L1	L2/FL language	Educational level
47	Elley and Mangubhai (1983)	FL	Fiji/Indian	0	English	Primary
48	Assi (2016)	L2	Arabic	1	English	Secondary
49	Mirzaei (2012)	FL	Persian	1	English	Other
50	Mermelstein (2014)	FL	Mandarin Chinese	1	English	Tertiary
51	Mermelstein (2015)	FL	Mandarin Chinese	1	English	Tertiary
52	Meganathan et al. (2019)	FL	89% Gamil L1, 7% English, 4% other	0	English	Primary
53	Meganathan et al. (2019)	FL	89% Gamil L1, 7% English, 4% other	0	English	Primary
54	Savasci & Akyel (2022)	FL	Turkish	1	English	Tertiary
55	O'Neill (2012)	FL	Japanese	1	English	Tertiary
56	Lituanas et al. (2001)	FL	Filipino, Cebuano	0	English	Secondary
57	French (1995)	L2	Chinese, Japanese, Korean, Indonesian, Spanish, Portuguese, Arabic	0	English	Tertiary
58	Rezaee et al. (2021)	FL	Persian	1	English	Tertiary
59	Rezaee et al. (2021)	FL	Persian	1	English	Tertiary
60	Tan and Kasuma (2019)	FL	Malay	1	English	Tertiary
61	Abdullah et al. (2012)	FL	Malay	1	English	Tertiary
62	Aka (2019)	FL	Japanese	1	English	Secondary
63	Huffman (2014)	FL	Japanese	1	English	Tertiary
64	Lee (2007)	FL	-	1	English	Tertiary
65	Lee (2007)	FL	-	1	English	Tertiary
66	Lee (2007)	FL	Mandarin Chinese	1	English	Tertiary
67	Javid and Al-Khairi (2011)	FL	Arabic	1	English	Tertiary
68	Yang et al. (2021)	FL	Mandarin Chinese	1	English	Secondary
69	Hamada (2020)	FL	Japanese	1	English	Tertiary
70	Alavi and Keyvanshekouh (2012)	FL	Persian	1	English	Tertiary

Table 5 (continued)

No	Authors (year)	L2/FL	L1 language	Same L1	L2/FL language	Educational level
71	Bell (2001)	FL	Arabic	1	English	Other
72	Alahirsh (2014)	FL	Arabic	1	English	Tertiary
73	Hsu and Lee (2009)	FL	Mandarin Chinese	1	English	Secondary
74	Jeon (2008)	FL	Korean	1	English	Tertiary
75	Lai (1993)	FL	Mandarin Chinese	1	English	Secondary
76	Matsui and Noro (2010)	FL	Japanese	1	English	Secondary
77	Kim and Ro (2023)	FL	South Korean	1	English	Primary
78	Iwata (2022)	FL	Japanese	1	English	Tertiary
79	Shimono (2019, 2023)	FL	Japanese	1	English	Tertiary
80	Nguyen and Baker (2023)	FL	Vietnamese	1	English	Primary
81	Misikir & Simegn (2025)	FL	Ethiopian	1	English	Secondary
82	Samatkao & Chaturongakul (2025)	FL	Thai	1	English	Secondary

Table 6 Intervention characteristics

No	Authors (year)	Researcher	Text limitation	Text selection	Account-ability	Help/instruction
1	Martinez (2017)	No	No	Yes	Yes	Yes
2	Yamamoto (2011)	-	Yes	No	Yes	Yes
3	Al-Jarf (2021)	Yes	No	No	Yes	No
4	Milliner (2019)	Yes	No	No	Yes	Yes
5	Tsang (1996)	No	No	No	Yes	No
6	Park et al. (2018)	No	No	No	No	No
7	Lake (2014)	No	No	No	No	No
8	Taguchi et al. (2004)	-	No	No	No	No
9	Lee and Lee (2023)	No	No	No	Yes	No
10	Huffman (2021)	Yes	Yes	No	Yes	Yes
11	Rezaee and Nourzadeh (2011)	No	No	No	Yes	No
12	Reynolds and Bai (2013)	-	No	No	No	No
13	Goto (2021)	Yes	No	No	No	Yes
14	Kara (2019)	-	Yes	No	No	No
15	Jafarpour (2014)	No	No	No	Yes	Yes
16	Naghbi and Moghaddam (2013)	Yes	No	No	No	No
17	Beglar et al. (2012)	No	No	No	Yes	Yes
18	Beglar et al. (2012)	No	No	No	Yes	Yes
19	Fiedler Vierma (1991)	Yes	No	No	Yes	No
20	Ahour and Zarei (2016)	No	No	No	Yes	No
21	Vaezi and Nilforooshan (2013)	-	Yes	No	Yes	No
22	Robb and Kano (2013)	No	Yes	No	Yes	No
23	McLean and Rouault (2017)	Yes	No	No	Yes	Yes
24	Rodrigo et al. (2004)	Yes	No	Yes	Yes	Yes
25	Park and Lee (2021)	Yes	No	No	Yes	No

Table 6 (continued)

No	Authors (year)	Researcher	Text limitation	Text selection	Account-ability	Help/instruction
26	Park and Lee (2021)	Yes	No	No	Yes	No
27	Burrows and Holsworth (2016)	Yes	Yes	No	No	No
28	Burrows (2012)	Yes	Yes	No	Yes	No
29	Lee et al. (2015)	No	No	Yes	Yes	Yes
30	Suk (2016)	Yes	No	Yes	Yes	Yes
31	Yilmaz et al. (2020)	Yes	Yes	No	Yes	Yes
32	Ro and Kim (2022)	-	No	No	Yes	No
33	Chen et al. (2013)	-	No	No	Yes	No
34	Schon et al. (1985)	No	No	No	No	No
35	Song and Sardegna (2014)	-	No	No	Yes	No
36	Lee and Schallert (2015)	No	No	No	Yes	Yes
37	Al-Homoud and Schmitt (2009)	Yes	Yes	No	No	No
38	Mason and Krashen (1997)	Yes	No	No	Yes	Yes
39	Mason and Krashen (1997)	Yes	No	No	Yes	No
40	Mason and Krashen (1997)	-	No	No	Yes	No
41	Modirkhamene and Gowrki (2011)	Yes	No	No	Yes	No
42	Sheu (2003)	No	No	No	No	Yes
43	Sheu (2003)	No	No	No	No	Yes
44	Fernandez de Morgado (2009)	Yes	No	No	Yes	No
45	Iwata (2020)	No	No	No	Yes	No
46	Shih (2015)	Yes	No	Yes	Yes	No
47	Elley and Mangubhai (1983)	No	No	No	No	No
48	Assi (2016)	No	No	No	No	No
49	Mirzaei (2012)	Yes	No	No	No	Yes
50	Mermelstein (2014)	Yes	Yes	No	No	No

Table 6 (continued)

No	Authors (year)	Researcher	Text limitation	Text selection	Account-ability	Help/instruction
51	Mermelstein (2015)	Yes	Yes	No	No	No
52	Meganathan et al. (2019)	No	Yes	No	Yes	No
53	Meganathan et al. (2019)	No	Yes	No	Yes	Yes
54	Savasci & Akyel (2022)	Yes	Yes	Yes	Yes	No
55	O'Neill (2012)	No	Yes	No	Yes	No
56	Lituanas et al. (2001)	Yes	Yes	Yes	Yes	Yes
57	French (1995)	Yes	Yes	No	Yes	No
58	Rezaee et al. (2021)	Yes	No	No	Yes	Yes
59	Rezaee et al. (2021)	Yes	No	No	Yes	Yes
60	Tan and Kasuma (2019)	-	No	Yes	No	No
61	Abdullah et al. (2012)	-	No	No	Yes	Yes
62	Aka (2019)	-	Yes	No	No	No
63	Huffman (2014)	Yes	No	No	Yes	No
64	Lee (2007)	Yes	No	No	Yes	No
65	Lee (2007)	Yes	No	No	Yes	No
66	Lee (2007)	Yes	No	No	Yes	No
67	Javid and Al-Khairi (2011)	Yes	No	No	Yes	Yes
68	Yang et al. (2021)	Yes	Yes	No	Yes	Yes
69	Hamada (2020)	No	Yes	No	Yes	Yes
70	Alavi and Keyvanshekouh (2012)	-	Yes	No	Yes	No
71	Bell (2001)	Yes	No	No	Yes	No
72	Alahirsh (2014)	No	Yes	No	Yes	No
73	Hsu and Lee (2009)	Yes	No	No	No	No
74	Jeon (2008)	Yes	No	No	Yes	Yes
75	Lai (1993)	No	Yes	No	Yes	No

Table 6 (continued)

No	Authors (year)	Researcher	Text limitation	Text selection	Account-ability	Help/instruction
76	Matsui and Noro (2010)	-	No	No	No	No
77	Kim and Ro (2023)	No	No	No	No	No
78	Iwata (2022)	No	No	No	Yes	No
79	Shimono (2019, 2023)	No	Yes	No	Yes	No
80	Nguyen and Baker (2023)	No	No	No	No	Yes
81	Misikir & Simegn (2025)	No	Yes	No	Yes	Yes
82	Samatkao & Chaturongakul (2025)	No	No	Yes	Yes	Yes

Table 7 Intervention characteristics (continued from Table 6)

No	Authors (year)	Interaction	Reward	Digital support program	Home activities	Home books	Graded readers	Text medium
1	Martinez (2017)	No	No	No	No	Yes	Yes	Print
2	Yamamoto (2011)	No	No	No	Yes	Yes	Yes	Print
3	Al-Jarf (2021)	Yes	No	No	Yes	Yes	-	Print
4	Milliner (2019)	Yes	Yes	Yes	Yes	Yes	Yes	Digital
5	Tsang (1996)	No	Yes	No	Yes	Yes	Yes	Print
6	Park et al. (2018)	No	No	No	Yes	Yes	Yes	Print
7	Lake (2014)	No	No	No	No	No	Yes	Print
8	Taguchi et al. (2004)	No	No	No	No	No	Yes	Print
9	Lee and Lee (2023)	No	No	No	No	No	-	Print
10	Huffman (2021)	Yes	Yes	Yes	No	No	Yes	Digital
11	Rezaee and Nourzadeh (2011)	No	No	No	Yes	Yes	Yes	Print
12	Reynolds and Bai (2013)	No	No	Yes	No	No	No	Digital
13	Goto (2021)	No	No	No	Yes	Yes	Yes	Print
14	Kara (2019)	No	No	No	Yes	Yes	Yes	Print
15	Jafarpour (2014)	No	No	No	Yes	Yes	Yes	Print
16	Naghbi and Moghaddam (2013)	No	No	No	No	No	-	Print
17	Beglar et al. (2012)	Yes	No	No	Yes	Yes	Yes	Print
18	Beglar et al. (2012)	Yes	No	No	Yes	Yes	Yes	Print
19	Fiedler Vierma (1991)	No	No	No	No	No	No	Print
20	Ahour and Zarei (2016)	No	No	No	Yes	Yes	Yes	Print
21	Vaezi and Nilforooshan (2013)	No	No	Yes	No	No	Yes	Digital
22	Robb and Kano (2013)	No	Yes	Yes	Yes	Yes	Yes	Digital
23	McLean and Rouault (2017)	No	Yes	Yes	Yes	Yes	Yes	Print
24	Rodrigo et al. (2004)	No	Yes	No	No	No	Yes	Print

Table 7 (continued)

No	Authors (year)	Interaction	Reward	Digital support program	Home activities	Home books	Graded readers	Text medium
25	Park and Lee (2021)	No	No	No	No	Yes	-	Digital
26	Park and Lee (2021)	No	No	No	No	Yes	-	Print
27	Burrows and Holsworth (2016)	No	No	No	No	No	Yes	Print
28	Burrows (2012)	No	Yes	No	Yes	Yes	Yes	Print
29	Lee et al. (2015)	No	No	No	No	No	Yes	Print
30	Suk (2016)	Yes	Yes	Yes	Yes	Yes	Yes	Print
31	Yilmaz et al. (2020)	Yes	No	No	Yes	Yes	Yes	Print
32	Ro and Kim (2022)	No	No	No	No	No	Yes	Print
33	Chen et al. (2013)	No	No	No	Yes	Yes	No	Digital
34	Schon et al. (1985)	No	No	No	No	No	No	Print
35	Song and Sardegna (2014)	Yes	No	No	No	No	Yes	Print
36	Lee and Schallert (2015)	Yes	No	No	No	No	-	Print
37	Al-Homoud and Schmitt (2009)	No	No	No	No	No	Yes	Print
38	Mason and Krashen (1997)	No	Yes	No	Yes	Yes	Yes	Print
39	Mason and Krashen (1997)	Yes	Yes	No	No	No	Yes	Print
40	Mason and Krashen (1997)	No	Yes	No	Yes	Yes	-	Print
41	Modirkhamene and Gowri (2011)	Yes	No	No	Yes	Yes	Yes	Print
42	Sheu (2003)	No	No	No	No	No	Yes	Print
43	Sheu (2003)	No	No	No	No	No	Yes	Print
44	Fernandez de Morgado (2009)	No	Yes	No	Yes	Yes	No	Print
45	Iwata (2020)	No	No	Yes	Yes	Yes	Yes	Print
46	Shih (2015)	No	No	No	Yes	Yes	Yes	Print
47	Elley and Mangubhai (1983)	No	No	No	No	No	-	Print
48	Assi (2016)	No	No	No	No	Yes	No	Print

Table 7 (continued)

No	Authors (year)	Interaction	Reward	Digital support program	Home activities	Home books	Graded readers	Text medium
49	Mirzaii (2012)	No	No	No	No	No	-	Print
50	Mermelstein (2014)	No	No	No	Yes	Yes	Yes	Print
51	Mermelstein (2015)	No	No	No	Yes	Yes	Yes	Print
52	Meganathan et al. (2019)	Yes	No	No	Yes	Yes	Yes	Print
53	Meganathan et al. (2019)	Yes	No	No	Yes	Yes	Yes	Print
54	Savasci & Akyl (2022)	Yes	No	No	No	No	Yes	Print
55	O'Neill (2012)	No	Yes	No	Yes	Yes	Yes	Print
56	Lituanas et al. (2001)	Yes	No	No	Yes	Yes	-	Print
57	French (1995)	No	Yes	No	No	No	Yes	Print
58	Rezaee et al. (2021)	Yes	No	No	Yes	Yes	Yes	Print
59	Rezaee et al. (2021)	Yes	No	No	Yes	Yes	Yes	Print
60	Tan and Kasuma (2019)	No	No	No	No	No	No	Digital
61	Abdullah et al. (2012)	Yes	No	No	Yes	Yes	-	Print
62	Aka (2019)	No	No	No	No	No	-	Print
63	Huffman (2014)	Yes	Yes	No	No	Yes	Yes	Print
64	Lee (2007)	No	No	No	Yes	Yes	Yes	Print
65	Lee (2007)	Yes	Yes	No	Yes	Yes	Yes	Print
66	Lee (2007)	Yes	Yes	No	No	No	Yes	Print
67	Javid and Al-Khairi (2011)	No	No	No	No	No	No	Print
68	Yang et al. (2021)	Yes	No	No	Yes	Yes	Yes	Print
69	Hamada (2020)	No	No	No	No	No	No	Print
70	Alavi and Keyvanshekouh (2012)	No	Yes	Yes	No	No	Yes	Print
71	Bell (2001)	No	No	No	Yes	Yes	Yes	Print
72	Alahirsh (2014)	No	No	No	No	No	Yes	Print

Table 7 (continued)

No	Authors (year)	Interaction	Reward	Digital support program	Home activities	Home books	Graded readers	Text medium
73	Hsu and Lee (2009)	No	No	No	No	No	Yes	Print
74	Jeon (2008)	No	Yes	No	Yes	Yes	Yes	Print
75	Lai (1993)	No	No	No	Yes	Yes	-	Print
76	Matsui and Noro (2010)	No	No	No	No	No	Yes	Print
77	Kim and Ro (2023)	No	No	No	No	No	Yes	Print
78	Iwata (2022)	No	Yes	Yes	Yes	Yes	Yes	Print
79	Shimono (2019, 2023)	Yes	Yes	Yes	Yes	Yes	Yes	Print
80	Nguyen and Baker (2023)	Yes	No	No	Yes	Yes	Yes	Print
81	Misikir & Simegn (2025)	Yes	No	No	Yes	Yes	No	Print
82	Samatkao & Chaturongakul (2025)	Yes	No	No	Yes	Yes	Yes	Print

Table 8 Study characteristics

No	Authors (year)	Peer review	Design	Random assign-ment	Randomization level	Study quality
1	Martinez (2017)	Yes	Quasi	-	Class	1
2	Yamamoto (2011)	Yes	-	-	Individual	2
3	Al-Jarf (2021)	Yes	Quasi	-	Class	2
4	Milliner (2019)	Yes	Quasi	-	Class	2
5	Tsang (1996)	Yes	Quasi	Yes	Class	4
6	Park et al. (2018)	Yes	Quasi	-	Class	3
7	Lake (2014)	Yes	Quasi	-	Class	2
8	Taguchi et al. (2004)	Yes	Quasi	-	Individual	2
9	Lee and Lee (2023)	Yes	Quasi	Yes	Class	3
10	Huffman (2021)	No	Quasi	-	Individual	3
11	Rezaee and Nourza-deh (2011)	Yes	Quasi	Yes	Class	4
12	Reynolds and Bai (2013)	Yes	Experimental	Yes	Individual	3
13	Goto (2021)	Yes	-	-	-	2
14	Kara (2019)	Yes	Quasi	-	-	2
15	Jafarpour (2014)	Yes	Quasi	Yes	Class	3
16	Naghbi and Moghaddam (2013)	Yes	Experimental	Yes	Individual	3
17	Beglar et al. (2012)	Yes	Quasi	-	Class	3
18	Beglar et al. (2012)	Yes	Quasi	-	Class	3
19	Fiedler Vierma (1991)	No	Quasi	No	Class	1
20	Ahour and Zarei (2016)	Yes	Quasi	Yes	Class	3
21	Vaezi and Nilforooshan (2013)	Yes	Experimental	Yes	Individual	3
22	Robb and Kano (2013)	Yes	Quasi	No	Class	3
23	McLean and Rouault (2017)	Yes	Experimental	Yes	Individual	5
24	Rodrigo et al. (2004)	Yes	Quasi	-	Class	2
25	Park and Lee (2021)	Yes	Experimental	Yes	Class	5
26	Park and Lee (2021)	Yes	Experimental	Yes	Class	5
27	Burrows and Holsworth (2016)	Yes	Experimental	Yes	Class	5
28	Burrows (2012)	No	Quasi	No	Class	4
29	Lee et al. (2015)	Yes	Quasi	Yes	Class	4
30	Suk (2016)	No	Quasi	-	Class	3
31	Yilmaz et al. (2020)	Yes	Quasi	Yes	Class	3
32	Ro and Kim (2022)	Yes	Quasi	-	School	3
33	Chen et al. (2013)	Yes	Quasi	-	Class	2

Table 8 (continued)

No	Authors (year)	Peer review	Design	Random assignment	Randomization level	Study quality
34	Schon et al. (1985)	Yes	Quasi	-	Class	3
35	Song and Sardegna (2014)	Yes	Quasi	No	-	3
36	Lee and Schallert (2015)	Yes	Quasi	Yes	Class	5
37	Al-Homoud and Schmitt (2009)	Yes	Experimental	Yes	Individual	3
38	Mason and Krashen (1997)	Yes	Quasi	No	Class	2
39	Mason and Krashen (1997)	Yes	Quasi	No	Class	2
40	Mason and Krashen (1997)	Yes	Quasi	No	Class	2
41	Modirkhamene and Gowrki (2011)	Yes	Experimental	Yes	Individual	3
42	Sheu (2003)	Yes	Quasi	-	Class	3
43	Sheu (2003)	Yes	Quasi	-	Class	3
44	Fernandez de Morgado (2009)	Yes	Quasi	No	Class	2
45	Iwata (2020)	Yes	Quasi	No	Individual	2
46	Shih (2015)	Yes	Quasi	-	Class	3
47	Elley and Mangubhai (1983)	Yes	Quasi	Yes	School	3
48	Assi (2016)	No	Quasi	Yes	Class	0
49	Mirzaei (2012)	Yes	Quasi	-	Class	2
50	Mermelstein (2014)	Yes	Quasi	Yes	Class	5
51	Mermelstein (2015)	Yes	Quasi	Yes	Class	5
52	Meganathan et al. (2019)	Yes	Experimental	Yes	Individual	4
53	Meganathan et al. (2019)	Yes	Experimental	Yes	Individual	4
54	Savasci & Akyel (2022)	Yes	Experimental	Yes	Individual	4
55	O'Neill (2012)	Yes	Quasi	-	Class	5
56	Lituanas et al. (2001)	Yes	Experimental	Yes	Individual	1
57	French (1995)	Yes	Quasi	No	Class	2
58	Rezaee et al. (2021)	Yes	Quasi	-	Class	2
59	Rezaee et al. (2021)	Yes	Quasi	-	Class	2
60	Tan and Kasuma (2019)	Yes	Quasi	Yes	Class	4
61	Abdullah et al. (2012)	Yes	Quasi	Yes	Class	3
62	Aka (2019)	Yes	Quasi	No	Class	4

Table 8 (continued)

No	Authors (year)	Peer review	Design	Random assignment	Randomization level	Study quality
63	Huffman (2014)	Yes	Quasi	Yes	Class	5
64	Lee (2007)	Yes	Quasi	No	Class	1
65	Lee (2007)	Yes	Quasi	No	Class	1
66	Lee (2007)	Yes	Quasi	No	Class	1
67	Javid and Al-Khairi (2011)	Yes	Quasi	No	Class	2
68	Yang et al. (2021)	Yes	Quasi	No	Class	3
69	Hamada (2020)	Yes	Quasi	No	Class	2
70	Alavi and Keyvanshekouh (2012)	Yes	Experimental	Yes	Individual	4
71	Bell (2001)	Yes	Quasi	-	Class	3
72	Alahirsh (2014)	No	Experimental	Yes	Individual	3
73	Hsu and Lee (2009)	Yes	Quasi	-	Class	3
74	Jeon (2008)	Yes	Quasi	Yes	Class	5
75	Lai (1993)	Yes	Quasi	-	Class	2
76	Matsui and Noro (2010)	Yes	Quasi	-	Class	2
77	Kim and Ro (2023)	Yes	-	-	-	3
78	Iwata (2022)	Yes	-	No	-	3
79	Shimono (2019, 2023)	Yes	Quasi	-	-	3
80	Nguyen and Baker (2023)	Yes	Quasi	-	Class	3
81	Misikir & Simegn (2025)	Yes	Quasi	-	Class	4
82	Samatkao & Chatu-rongakul (2025)	Yes	Quasi	No	-	0

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Author Contribution The initial idea for the study was developed by Author 3 and 5. All authors contributed to the study conception and design. In the search for studies to-be-included in the meta-analysis, the first 500 titles (out of 1710 hits) were screened on title and abstract according to the in- and exclusion criteria by Author 1, 2, 4, and 5; the other 1210 titles were screened by Author 1 and 2. The coding scheme was developed by Author 1 and 2. A first version of the coding scheme was tried out on a sample of three studies by Author 1, 2, and 5. These authors coded the studies, compared their codes, and adjusted the scheme based on their comparisons. Subsequently, Author 1 and 2 coded the remaining studies of the initial search. In case of uncertainties, Author 5 was consulted for a final decision. Author 1 and 5 double-coded the six additional studies that were included based on the update of the search. All analyses were carried out by Author 1 and 5. The manuscript was written by Author 1 and 5.

Declarations

Conflict of interest The authors declare no competing interests.

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